GRASP: Gated Regression-Aware Skill Proposer

for Self-Improving LLM Agents · J. Moll, J.-P. Corbeil, J. Pan, M. Hadamitzky, D. Rueckert, L. Adams, K. Bressem · TUM · Microsoft HLS · arXiv 2605.29668v3 (CC BY 4.0)

HarnessRSI 84 ★★★

evidence: retained

Paper

Code

1 What changes, what is kept

The library: ≤ 10 skills, ≈ 5.6k tokens

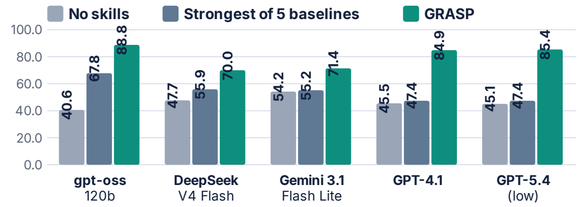
Trigger	when the skill applies	**Rule**	behaviour to follow
Verify	optional check	**Example**	contrastive case

1. **Run batch, group failures**
   48 dev episodes; failed traces labelled by failure mechanism
2. **Writer proposes K = 4 edits**
   each edit adds, modifies or removes one skill
3. **Probe every candidate**
   36 earlier examples, half failing / half passing; current library re-run too (F0, R0)
4. **Acceptance gate**
   (F−F0) − (R−R0) > 0 and R ≤ R0 · best passing edit applied, else library unchanged
5. **New library version**
   injected into the agent's context for the next batch; rejected edits leave no trace

5 ep.

best-validation checkpoint → test and OOD run once
weights never change; the kept artifact is the library

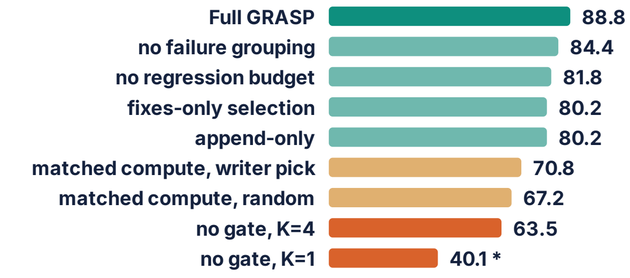
2 Headline: MedAgentBench test accuracy



Accuracy (%) vs FHIR server state; 5 seeds open models, 3 proprietary. gpt-oss: +21.0 over Evo-MedAgent, CI [10.3, 33.8], p = 0.008. OOD 56.3 vs 31.3: CI [3.7, 44.0], **p = 0.087, not significant at 0.05**; 3-seed p floor 0.10.

3 Controls that decide if the gain is real

Ablation · test (%), gpt-oss-120b (Table 4)



Give the baselines the same gate (Table 5, 5 seeds)

Gate granted to…	ungated T / OOD	+ gate K=4 T / OOD
Seq. memory	41.2 / 16.7	49.2 / 15.0
Batch memory	34.4 / 13.0	62.2 / 22.0
ExpeL	49.1 / 12.0	55.9 / 13.3
SkillX	53.1 / 21.3	55.9 / 19.3
Evo-MedAgent	67.8 / 31.3	**55.7 / 26.1**
GRASP	40.1 / 27.2	**88.8 / 56.3**

At equal K=1: GRASP 73.4 ± 23.1 / 50.9 ± 20.0 vs gated Evo 69.4 ± 9.2 / 30.7 ± 11.8 — in-domain gap inside noise; 88.8 needs K=4. K=4 lowers Evo (69.4 → 55.7) and SkillX (57.5 → 55.9). Gate lifts baselines in-domain (+1.6 to +15.0, several inside seed noise), none OOD. Batch memory: 439 probe calls vs 411; gpt-oss / MedAgentBench only. *No gate, K=1: test ≈ no skills (40.6); OOD 27.2 vs 8.7.

4 Transfer — and where the gain is absent

- **GPT-5.4 library → gpt-oss:** OOD 77.8 vs 56.3 own, in-domain 76.0 vs 88.8. Weaker-writer libraries are mostly worse (exceptions; Gemini cells SD > 30). Baselines compared are **ungated**, 2 configs only; their GPT-5.4 libraries are weak at source (OOD 1.7–16.7)
- **MedAgentBench-v2 OOD, below no skills:** DeepSeek 67.0 vs 83.6, GPT-4.1 82.8 vs 84.9, GPT-5.4 80.0 vs 82.7; gpt-oss 86.0 vs 87.1 (SkillX) — "no superiority claim"
- **v2 in-domain:** GPT-4.1 70.3 vs no skills 70.5; GPT-5.4 63.3 ≈ best baseline; Gemini 67.2 vs 65.1 within noise
- **FHIR-AgentBench:** self-training 51.5 → 58.2 only; libraries from other benchmarks (47.2 / 45.7 / 48.3) fall below baseline
- **Non-clinical (exploratory, gpt-oss, 3 seeds, vs no skills only, 20–60 test episodes):** OS Interaction 48.6 → 49.5, no gain; ALFWorld 23.3 → 51.7

Takeaway

Gating each skill edit on a held-out probe with a regression budget is what drives the gain: with no gate at K=1, test accuracy falls to the no-skills level. The gain is large on MedAgentBench, smaller on FHIR-AgentBench, mixed on v2, flat on OS Interaction.

Our read · HarnessRSI

Score 84 · ★★★ · evidence level: retained

Reports: held-out + OOD, transfer, 3–5 seeds, ablations, gate granted to baselines, leakage audit of 106 libraries (App. F).
Not shown: gated-baseline transfer; gate-for-baselines beyond one model. Improver is compared, so level may be understated.